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SYNC

ENTERPRISE TRACK A · PROJECT UDYAM DIGITAL

# Udyam Digital

A small business goes live in one sitting: on a phone, in Hindi or English, with no salesperson in the loop.



**Live prototype** udyam-digital-isyfatadrab4adtbqddzwp.streamlit.app
**Code** github.com/Sheel23501/udyam-digital

Sheel Gautam · IIIT Delhi · November 2026

Platform Partner unstop

# ₹35.6 lakh of ₹55.4 lakh went to leads that never activated

4,632 of 5,000 leads dropped last quarter. Half of those drops were the sales process itself, not the offer.

**48%**

of drops: unreachable, no time for a visit, visit not scheduled, quote too slow or too complex

**₹53,026**

field-sales cost per activation on micro retail, for a ₹2,619/month customer

**17 days**

median lead-to-activation; the quote alone takes 4 days

**719 → 9**

owners asked for 5G AirFiber 719 times; sales sold it 9 times

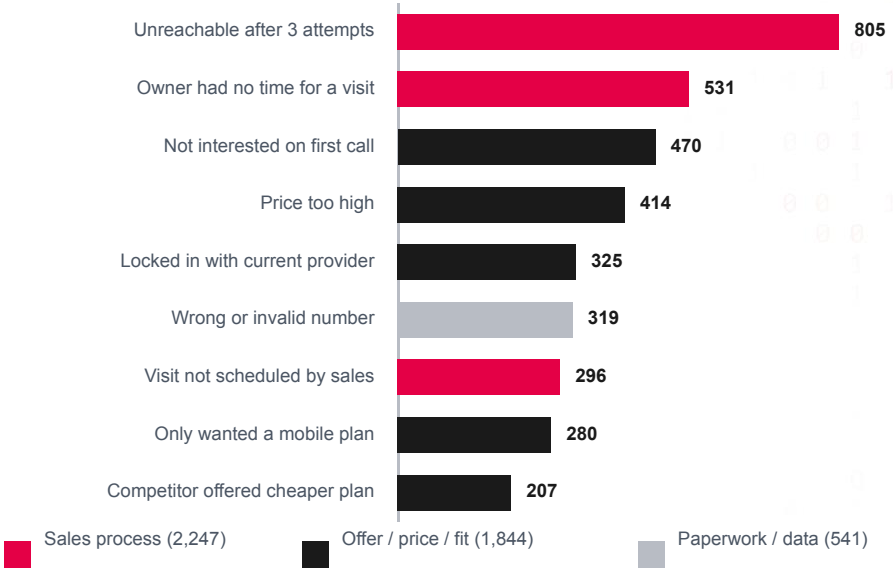
**22**

customers activated last quarter with a cancelled or suspended GSTIN

**6% → 28%**

6-month churn as the status calls a customer had to make rose from 1 to 5+

**Why 4,632 leads dropped (top 9 reasons)**



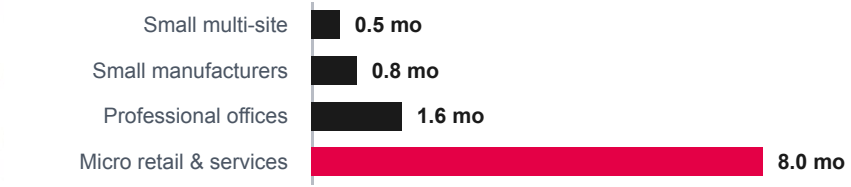

Source: smb_leads.csv, 5,000 leads (2026-04-01 to 2026-06-30); every figure reproduced by backend/scripts/make_charts.py.

# Where to go first: payback decides

Blended CAC per activation (all leads' acquisition cost ÷ activations) against the monthly bundle value, by segment.

| Segment | Leads | Activation | ₹/month | 6-mo churn | CAC / activation | Payback | Our move |
|---|---|---|---|---|---|---|---|
| **Small multi-site** | 755 (15%) | 11.0% | ₹31,233 | 11% | ₹15,614 | 0.5 mo | Wave 1: digital discovery, field close |
| **Small manufacturers** | 289 (6%) | 12.8% | ₹19,980 | 8% | ₹15,324 | 0.8 mo | Wave 2: field-assisted (ILL, SD-WAN) |
| **Professional offices** | 1,247 (25%) | 11.1% | ₹6,796 | 14% | ₹10,695 | 1.6 mo | Wave 1: digital, optional assist |
| **Micro retail & services** | 2,709 (54%) | 4.1% | ₹2,505 | 23% | ₹20,007 (field ₹53,026) | 8.0 mo | Digital only; no field visits |

**Months of revenue to repay the acquisition cost**




**Verdict.** Professional offices and small multi-site first: they pay back in weeks. Micro retail goes digital-only, where a field visit costs ₹53,026 to win ₹2,619 a month. Manufacturers come in wave 2 with field assistance for leased lines and SD-WAN.

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# One bundle per segment, one monthly number

Composed by the rules engine from the catalogue's eligibility rules and minimum prices; the owner can switch items on and off and the price re-validates instantly.

## Dukaan Start

Micro retail · GST-free path

from **₹1,198** /month

Setup ₹0 · 6-month · live in 2 days

- Biz 5G AirFiber 100
- Business Postpaid 299

*Live in 2 days, no setup fee, no GST needed. For the 1,466 no-GST leads (2.0% activation today).*

## Clinic / Office Pro

Professional offices

from **₹5,487** /month

Setup ₹1,500 · 12-month · live in 5 days

- Biz Fibre 300
- Business Postpaid 599 ×2
- Cloud Email & Office Suite ×8
- Cloud Backup 1 TB
- Endpoint Security ×8
- Backup 4G/5G Failover

*Email for every user, backup, endpoint security and a failover line, because a clinic cannot be offline.*

## Multi-Store

Small multi-site · 3 outlets

from **₹21,579** /month

Setup ₹24,800 · 24-month · live in 14 days

- Biz Fibre 300 ×3
- Business Postpaid 599 ×6
- Backup 4G/5G Failover ×3
- Managed SD-WAN ×3
- Toll-free Number
- Cloud CCTV ×6
- Cloud POS & Billing ×3

*Every outlet on one network with cameras and billing that never stops; SD-WAN and toll-free need GST.*

## Factory Connect

Small manufacturers

from **₹14,377** /month

Setup ₹17,000 · 24-month · live in 21 days

- Internet Leased Line 20
- Business Postpaid 599 ×3
- Machine Monitoring IoT ×6
- Endpoint Security ×12
- Cloud Backup 1 TB
- Managed Firewall

*A leased line with an SLA, machine monitoring and a managed firewall; 21-day install stated up front.*

Prices are catalogue minimums composed by backend/app/engine (python scripts/run_scenarios.py reproduces them); ranges and setup fees shown to the owner on the quote screen.

# The journey: six steps on a phone, three rules underneath

Hindi, English or Hinglish, detected from what the owner types. Works even when the model is down: the engine still quotes.

**1 Enter**

Link, QR or WhatsApp. Mobile number + one-time code.

**2 Advisor**

4 to 6 questions in the owner's language. Facts recorded, never products.

**3 Quote**

One monthly number. Toggle items; every change re-checked by the rules.

**4 Consent**

Itemised DPDP notice in their language before any document.

**5 Smart KYC**

Four photos. 12 checks against GST and Udyam registries in seconds.

**6 Tracker**

Dated stages, notifications. Zero status calls.

## Rules decide, the LLM narrates

A deterministic engine over the catalogue decides eligibility, dependencies and price. The model only asks questions and explains. The card is drawn from engine output, never from prose. 30 scenario fixtures and a 900-profile invariant test keep it honest.

## One number, instantly

No PDF, no call-back: the quote appears in the chat and re-validates on every toggle. Owners who heard 'quote too complex' or 'quote took too long' (385 drops) get a price in minutes.

## No status calls

Every order has dated stages and proactive notifications. Customers who made one status call churned at 6%; five calls, 28%. The tracker exists to make that number zero.

Architecture diagram and data flows: docs/architecture.png (submitted separately). Prototype: FastAPI + rules engine + Streamlit, LLM via an OpenAI-compatible provider chain.

# Smart KYC: 12 checks in seconds, built around the DPDP Act

Vision extraction of four documents, then deterministic validators against the GST and Udyam registries. A photo problem is a REVIEW, never a rejection.

**The 12 sample applicants as the prototype grades them: 12/12 match the key**

| | | |
|---|---|---|
| FAIL **K01** GSTIN cancelled; Udyam number missing | REVIEW **K02** GST certificate photo rotated | PASS **K03** all documents consistent |
| REVIEW **K04** GST certificate photo blurred | FAIL **K05** Udyam certificate missing | FAIL **K06** GSTIN check digit fails: tampered digit |
| FAIL **K07** legal name differs across documents | PASS **K08** all documents consistent | PASS **K09** all documents consistent |
| FAIL **K10** signatory is not the person of record | FAIL **K11** electricity bill 257 days old (limit 90) | PASS **K12** all documents consistent |

**22 → 0**

activations with a cancelled or suspended GSTIN: the registry is checked live, before the order

## DPDP controls, each a code path, not a slide

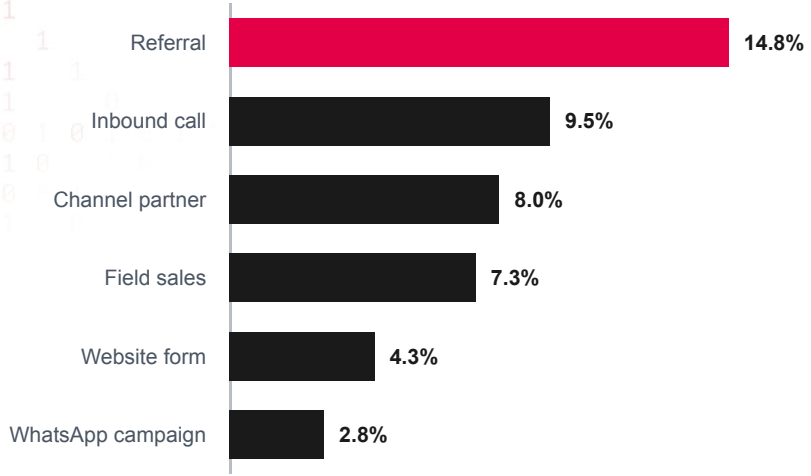
**Consent gate:** documents are refused (HTTP 403) until the owner ticks four itemised purposes in their language
**Minimisation:** five facts for the quote; GST or Udyam path; no Aadhaar; only the image leaves the boundary
**Masking:** GSTIN as 27XXXXXXXXXX1ZP, names as 'Nisha A.' in UI, API and audit log
**Retention:** images deleted 30 days after a rejection, 90 after verification; erasure on request
**Accuracy:** REVIEW before any adverse decision; 'talk to a human' always available

Reproduce: cd backend && python scripts/kyc_report.py --key tests/kyc_answer_key_images.json. Privacy note: docs/privacy-note.md (submitted separately).

# Digital adds; the field force multiplies

Take the field team out of the ₹2,500 kirana sale and put it where payback is under a month. Attribution keeps partners on side.

### Activation rate by lead source




*Referral converts at 14.8% but churns at 24.6%: pay the referral bonus at month 3, not at activation.*

### Micro retail goes digital-only

Field sales spent ₹53,026 per activation here. Removing those 27 visits a quarter saves about ₹14 lakh per quarter with no loss of volume.

### Field team redeployed to multi-site and manufacturers

Payback 0.5 and 0.8 months. The advisor qualifies and quotes; a human closes the ₹20,000-plus accounts and handles ILL and SD-WAN installs.

### Partners keep their code

Every journey carries a partner code for attribution and commission; partners see the scored queue for their area.

### Nothing dead-ends

Leads the journey cannot close land in a back-office queue ranked by a propensity model (holdout AUC 0.68) plus how recoverable the drop reason is, each with a suggested next action.

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# Rollout in three phases

Start where the data says the money is, with the adapters that make mocks real.

**Phase 0 · Q3 2026**

## Prove it

- Delhi + Mumbai
- Hindi and English, Hinglish auto-detect
- Micro retail (GST-free) + professional offices
- Web and WhatsApp entry, partner-code pilot with 20 partners
- Mock registries swapped for GSTN/GSP and Udyam APIs

**Gate: 12% blended activation, quote in under 5 minutes, zero activations with a non-active GSTIN**

**Phase 1 · Q4 2026**

## Widen

- Tier-2 cities (Jaipur, Lucknow, Indore, Nagpur, Surat...)
- Small multi-site journeys with per-site quotes
- DigiLocker pull of GST/Udyam certificates, eSign for the order
- Field team re-routed by the propensity queue

**Gate: CAC per activation ≤ ₹7,500; KYC first-pass ≥ 70%**

**Phase 2 · Q1 2027**

## Deepen

- Manufacturers: leased line and SD-WAN journeys with field assist
- Real-time fibre serviceability at quote time
- 30-day check-in and referral bonus at month 3
- Regional languages beyond Hindi where volume justifies it

**Gate: 6-month churn ≤ 10%; status calls per customer < 0.5**

Dependencies: GSTN/GSP and Udyam API access, DigiLocker and eSign onboarding, a DLT-registered SMS template for OTPs, partner commission rules.

# What we will measure

Q2 2026 baseline from the leads file; two-quarter targets, and the design feature that moves each one.

| Metric | Today | Target (2 quarters) | What moves it |
|---|---|---|---|
| **Lead → activation** | 7.4% | 12% blended; 15% for offices and multi-site | Instant quote, GST-free path, no visit-dependent steps |
| **Lead → activation time** | 17 days (quote alone 4) | ≤ 5 days, install-bound; quote in < 5 minutes | Advisor + engine + instant KYC |
| **Status calls per customer** | 3.0 | < 0.5 | Tracker with dated stages and proactive notifications |
| **Blended CAC per activation** | ₹15,053 | ≤ ₹7,500 (micro ≤ ₹3,000) | Digital-only micro, referral, redeployed field force |
| **6-month churn** | 15.2% | ≤ 10% | Fewer calls, right-sized bundles, 30-day check-in, bonus at month 3 |
| **Activations with non-active GSTIN** | 22 | 0 | Live registry check inside KYC, before the order |
| **KYC first-pass rate** | n/a | ≥ 70%; REVIEW resolved within 24 h | Human-language fixes, re-upload path, Udyam-only path |

*Measured weekly from the same tables the prototype already writes: sessions, consents, kyc_applications, orders, audit_log.*

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# Risks we designed for, and what is already built

Each mitigation is a code path in the prototype, not a plan.

| Risk | Mitigation in the prototype |
|---|---|
| **The model recommends something the owner is not eligible for** | The rules engine decides every item; the model only narrates; the UI renders engine JSON. 30 scenario fixtures + 16 live transcripts, zero violations |
| **KYC wrongly rejects a genuine owner** | Photo problems are REVIEW with a plain-language fix; FAIL always carries the reason and a 'talk to a human' hand-off |
| **LLM or vision API down or over quota during a demo** | Provider chain with fallbacks; cached demo replies; the engine and validators run offline. The journey completes with no model at all |
| **Fibre not feasible, install delays** | 5G AirFiber as a 2-day interim line; install days stated on the quote; tracker dates |
| **Channel conflict with partners and field sales** | Partner code on every journey; propensity queue routes high-value leads to humans; micro retail digital-only |
| **Privacy and DPDP exposure** | Consent gate (403), masking everywhere, retention purge, erasure, staff-only back office, audit log |
| **Low-quality leads from digital channels** | Mobile number verified by one-time code at entry; GST status checked live |

## Built and verified

**2,050** automated tests, all green in CI

**30 / 30** advisor scenarios through the rules engine

**12 / 12** KYC verdicts matching the answer key

**16 / 16** live conversations on Gemini, zero rule violations

**Docker** compose build and smoke test on every push

**Try it:** the README's step-by-step guide takes a judge from clone to a verified API in 20 minutes, no keys needed.

Repository: github.com/Sheel23501/udyam-digital · docs/how-to-check.md · docs/privacy-note.md · docs/architecture.png